



WSQ Courseware Generator

Complete User Guide

Step-by-Step Instructions for All Features

For New Team Members & Interns

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Prepared by: Tertiary Infotech Academy Pte Ltd

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1. Introduction — What is WSQ?

1.1 About WSQ (Workforce Skills Qualifications)

The **Singapore Workforce Skills Qualifications (WSQ)** system is a national credential system that trains, develops, assesses, and certifies skills and competencies of the Singapore workforce. WSQ is developed by SkillsFuture Singapore (SSG) together with industry partners.

Training providers (like Tertiary Infotech Academy) must submit a standardised set of courseware documents to SSG for each WSQ course they offer. These documents must follow strict formatting and content requirements. This is where the WSQ Courseware Generator comes in — it automates the creation of these documents.

In simple terms: When a company wants to offer a WSQ-certified course, they first get a Course Proposal (CP) approved by SSG. Then they need to create a full package of training documents — lesson plans, assessment papers, teaching guides, etc. This app generates all of those documents automatically from just the CP.

1.2 What This App Does

The **WSQ Courseware Generator** is an AI-powered web application that takes a single **Course Proposal (CP)** document and generates **all required courseware** automatically. Instead of manually creating 8+ documents for each course (which can take days), the app produces them in minutes.

The app generates the following documents:

- **Assessment Plan (AP)** — Defines how trainees will be assessed, what methods are used, and what evidence is required
- **Assessment Summary Record (ASR)** — A form for recording each trainee's assessment results (pass/fail)
- **Facilitator Guide (FG)** — The instructor's teaching manual with lesson content, activities, examples, and trainer notes
- **Learner Guide (LG)** — The student's handbook with learning content, exercises, and self-check questions
- **Lesson Plan (LP)** — Day-by-day training timetable (e.g., 9am-10am Topic 1, 10am-12pm Topic 2, etc.)
- **Assessment Papers** — Question papers and answer keys for up to 9 assessment types (written, practical, case study, etc.)
- **Slide Deck (PPTX)** — PowerPoint presentation slides with infographics for the trainer to use during class
- **Marketing Brochure (PDF)** — A professional PDF brochure for promoting the course to potential trainees

***Note:** You do not need to be a technical person to use this app. It runs in your web browser like any website. Just upload the CP, click a few buttons, and download the generated documents.*

1.3 Key Terminology (Glossary)

Term	Meaning
CP	Course Proposal — the approved document submitted to SSG describing a course
AP	Assessment Plan — document outlining assessment strategy and criteria
ASR	Assessment Summary Record — form for recording trainee assessment outcomes
FG	Facilitator Guide — instructor's teaching manual with lesson content
LG	Learner Guide — student's handbook with learning materials
LP	Lesson Plan — day-by-day timetable/schedule for the course
TGS	Training Grant for Skills — unique reference code for each WSQ course (e.g., TGS-2024001234)
TSC	Training & Skills Competency — the competency framework a course maps to
SSG	SkillsFuture Singapore — government agency that oversees WSQ
LU	Learning Unit — a major section of the course (a course has 1 or more LUs)
K statement	Knowledge statement — what the trainee should KNOW
A statement	Ability statement — what the trainee should be able to DO
UEN	Unique Entity Number — Singapore business registration number
SAQ / WA	Short Answer Questions / Written Assessment
PP	Practical Performance assessment
CS	Case Study assessment
OQ	Oral Questioning assessment
OI	Oral Interview assessment
DEM	Demonstration assessment
RP	Role Play assessment
PRJ	Project-based assessment
ASGN	Assignment-based assessment

2. Getting Started

2.1 Opening the App

The WSQ Courseware Generator runs as a web application in your browser — just like Google Docs or any website. No installation is needed.

Step 1: Open **Google Chrome** (recommended) or any web browser.

Step 2: Go to the **app link** provided to you by your administrator (e.g., `http://localhost:7860` or the public URL).

Step 3: The app will load with a dark-themed interface. You will see the **sidebar menu** on the left and the main content area on the right.

***Note:** You do not need to install Python, Claude, or any software. Everything runs in your browser. If the app link doesn't load, ask your administrator if the app is running.*

2.2 Selecting a Company

At the top of the left sidebar, you will see a **Company dropdown**. This determines which company's branding (logo, UEN, address) appears on all generated documents.

- Select the correct company **before** generating any documents.
- The default is 'Tertiary Infotech Academy Pte Ltd'.
- To add or edit companies, see **Section 12: Company Management**.

2.3 Understanding the Sidebar

The left sidebar contains the main navigation menu with 8 pages. Below the menu, you will find:

- **Companies button** — Opens company management settings.
- **Agent Status** — Shows if any AI background job is running.

***Note:** The recommended workflow follows the menu order from top to bottom: Extract Course Info → Generate AP/FG/LG → Lesson Plan → Assessment → Add Assessment to AP → Slides → Brochure → Audit.*

3. Extract Course Info

This is **always the first step** — you cannot skip it. You must extract course information from an approved Course Proposal (CP) before generating any documents. Think of it as loading the course data into the system. Once extracted, this data is automatically available on all other pages.

Tip: If you skip this step and go directly to Generate AP/FG/LG, you will see a warning: 'No course info loaded'. Always start here first.

3.1 Uploading a Course Proposal

Step 1: Navigate to **Extract Course Info** in the sidebar.

Step 2: Click the **file upload area** (drag-and-drop also works).

Step 3: Upload your Course Proposal file. Supported formats:

- **.docx** — Word document (most common)
- **.xlsx** — Excel spreadsheet

3.2 Entering Course Details

Step 1: Enter the **Course Reference Code (TGS)** — e.g., TGS-2024001234. This code appears on all generated documents.

Step 2: (Optional) Enter the **Course URL** — the link to the course webpage. This is only needed if you plan to generate a brochure later.

Step 3: Click the **Extract** button to start processing.

The system uses AI (Claude Agent) to interpret the Course Proposal and extract structured data. This runs in the background — you will see a **spinning indicator** while it processes. You can wait on this page or navigate away; the extraction will continue.

3.3 Reviewing Extracted Data

Once extraction completes, you will see a **success card** showing:

- Course Title
- Number of Learning Units (LUs)
- Number of Topics
- Assessment Methods

Below the success card, expandable tables show the full extracted data: course overview, learning outcomes, topics with bullet points, and assessment details. **Review this data carefully** — it forms the basis for all generated documents.

Tip: If the company name in the CP matches a company in the database, the sidebar company dropdown will auto-select it for you.

4. Generate AP / FG / LG

This page generates three core WSQ courseware documents using AI agents:

- **Assessment Plan (AP)** — Outlines assessment strategy, methods, and criteria.
- **Assessment Summary Record (ASR)** — Auto-generated alongside the AP.
- **Facilitator Guide (FG)** — Instructor manual with lesson content and activities.
- **Learner Guide (LG)** — Student handbook with learning content.

4.1 Selecting Documents to Generate

Step 1: Navigate to **Generate AP/FG/LG** in the sidebar.

Step 2: You will see the extracted course info summary at the top. If it says 'No course info loaded', go back to Step 3 first.

Step 3: Use the **checkboxes** to select which documents to generate:

- Learning Guide (LG) — checked by default
- Assessment Plan (AP) — checked by default
- Facilitator Guide (FG) — checked by default

***Note:** You can uncheck any document you don't need. For example, if you only need the LG.*

4.2 Running Generation

Step 1: Click the **Generate Documents** button.

Step 2: A spinner will appear as each document is being generated by the AI agent.

Step 3: Generation takes 1-3 minutes per document depending on course complexity.

The system automatically enriches the data with TSC (Training & Skills Competency) fields, proficiency levels, and sector information before generating.

4.3 Downloading Documents

Once generation completes, you have two download options:

- **Individual downloads** — Click the download button next to each document.
- **Bulk download** — Click **Download All as ZIP** to get all documents in one file.

Files are named as follows:

- LG_[TGSCode]_[CourseName].docx
- Assessment_Plan_[TGSCode]_[CourseName].docx
- Assessment_Summary_Record_[TGSCode]_[CourseName].docx
- FG_[TGSCode]_[CourseName].docx

Tip: All generated documents are also automatically saved to the Courseware/ folder in the project directory.

5. Generate Lesson Plan

The Lesson Plan is generated **instantly** using a pure Python scheduling algorithm — no AI processing is needed. It creates a day-by-day timetable for the course.

5.1 Generating the Schedule

Step 1: Navigate to **Generate Lesson Plan** in the sidebar.

Step 2: The extracted course info is auto-loaded. Verify the course details at the top.

Step 3: Click **Generate Lesson Plan**.

The schedule follows these rules:

- Daily hours: **9:00 AM – 6:00 PM**
- Lunch break: **12:30 PM – 1:15 PM** (45 minutes, fixed)
- Assessment: **4:00 PM – 6:00 PM** on the last day only
- Topics are distributed evenly across available time
- Breaks fill all remaining gaps to ensure a full 9am–6pm day

5.2 Previewing & Downloading

After generation, you will see:

- **Statistics** — Total days, minutes per topic, instructional hours, assessment hours.
- **Day-by-day preview** — Expandable sections showing each day's timetable.
- **Download button** — Downloads as LP_[TGSCode]_[CourseName]_v1.docx

6. Generate Assessment

Generate assessment question papers in standard WSQ format. Supports 9 assessment types.

Code	Assessment Type	Description
WA (SAQ)	Written Assessment	Short answer questions testing knowledge
PP	Practical Performance	Hands-on task demonstration
CS	Case Study	Scenario-based analysis questions
OQ	Oral Questioning	Verbal Q&A assessment
OI	Oral Interview	Structured interview assessment
DEM	Demonstration	Live skill demonstration
RP	Role Play	Simulated scenario performance
PRJ	Project	Extended project-based assessment
ASGN	Assignment	Written assignment submission

6.1 Uploading Facilitator Guide (Optional)

You can optionally upload the Facilitator Guide (FG) to extract Knowledge (K) and Ability (A) statements. This helps the AI map assessment questions to specific K/A items.

Step 1: Navigate to **Generate Assessment** in the sidebar.

Step 2: (Optional) Upload the FG .docx file to extract K/A statements.

6.2 Selecting Assessment Types

Step 1: Use the **checkboxes** to select which assessment types to generate.

Step 2: Click **Generate Assessments**.

Step 3: The AI agent creates questions, expected answers, and K/A mappings for each type.

6.3 Downloading Assessments

- **Individual downloads** — Each assessment type has its own download button.
- **Bulk download** — Download all as assessments_[CourseName].zip

7. Add Assessment to AP

After generating assessment papers (Section 6), you need to attach them to the Assessment Plan as annexes. This step merges the assessment question papers and answer keys into the AP document with proper annex labelling.

7.1 Uploading the AP and Assessment Files

Step 1: Navigate to **Add Assessment to AP** in the sidebar.

Step 2: Upload the **Assessment Plan (AP)** .docx file generated in Section 4.

Step 3: Upload your **assessment files** — the question papers and answer keys generated in Section 6.

The system auto-detects the assessment types from the filenames. For each type, it creates annex sections (Annex A, Annex B, etc.) with proper headers:

- Annex A: QUESTION PAPER OF [Assessment Type] ASSESSMENT
- Annex B: SUGGESTED ANSWER TO [Assessment Type] ASSESSMENT QUESTIONS

7.2 Merging & Downloading

Step 1: Click **Merge** to combine the AP with all assessment annexes.

Step 2: Download the merged document. Page breaks are automatically inserted between each annex section.

Tip: Always review the merged document in Microsoft Word to verify the annex formatting and page breaks look correct.

8. Generate Slides

Generates a presentation slide deck with infographics for the course content. The system uses a multi-agent pipeline to create content-rich slides with visuals.

8.1 Generating Slides

Step 1: Navigate to **Generate Slides** in the sidebar.

Step 2: Ensure course info has been extracted (Section 2).

Step 3: Click **Generate Slides** to start the multi-agent pipeline.

The pipeline runs through 5 phases:

- **Phase 1: Research** — Researches topic content
- **Phase 2: Content** — Generates slide content blocks
- **Phase 3: Editor** — Refines and structures content
- **Phase 4: Infographic** — Creates visual infographics
- **Phase 5: Assembly** — Builds the final PPTX file

8.2 Downloading the Slide Deck

Once complete, use the **Download** button to save the PPTX presentation file. The slide deck includes company branding, WSQ logos, and topic infographics.

Course Duration	Target Slide Count
1-day course	100 slides
2-day course	140 slides
3-day course	210 slides
4-day course	250 slides

9. Generate Brochure

Creates a marketing brochure PDF for any WSQ course. There are **two modes** depending on whether the course is a Tertiary Infotech course or a client course.

9.1 Tertiary Courses (URL Only)

For Tertiary Infotech courses, the brochure is generated entirely from the course webpage:

- Step 1:** Navigate to **Generate Brochure** in the sidebar.
- Step 2:** Leave the **Client Course** upload area empty.
- Step 3:** Paste the **Course URL** (e.g., from MySkillsFuture or tertiarycourses.com).
- Step 4:** Click **Generate Brochure**.

The system scrapes all course details (title, topics, fees, requirements) from the URL and generates a branded PDF with the Tertiary Infotech logo and contact details.

9.2 Client Courses (Upload CP)

For client courses (e.g., Laures, other training providers), MySkillsFuture has limited info. Instead, upload the client's Course Proposal:

- Step 1:** Navigate to **Generate Brochure** in the sidebar.
- Step 2:** Upload the client's **Course Proposal** (.docx or .xlsx).
- Step 3:** Enter the **TGS Reference Code** (e.g., TGS-2026062163).
- Step 4:** (Optional) Paste the **Course URL** — used to fetch the estimated payable fee.
- Step 5:** Click **Generate Brochure**.

The system will automatically:

- **Extract course data** from the CP (title, topics, learning outcomes, description).
- **Detect the company name** from the CP and look up the company in the database.
- **Replace all branding** — company name, logo, address, email from the database.
- **Fetch the estimated fee** from the URL (if provided).
- **Remove Tertiary-specific content** — phone, WhatsApp, registration link, funding table.

Note: The client company must be added in Company Management (Section 12) with their address, email, and logo **BEFORE** generating the brochure. Otherwise the brochure will have missing details.

Tip: The company name matching is flexible — it ignores uppercase/lowercase, periods, and extra spaces. So 'LAURES SOLUTIONS PTE. LTD.' in the CP will correctly match 'Laures Solutions Pte Ltd' in the database. You do not need to worry about exact formatting.

9.3 Downloading & Clearing

- **Download PDF** — Click to save the brochure.
- **Clear & Generate New** — Click to reset and upload a new CP for the next course.

Tip: When generating brochures for multiple client courses, use the **Clear** button after each one to ensure fresh data for the next course.

10. Convert Documents

Convert existing (non-standard) documents into standardized WSQ format. This page has **three tabs**:

10.1 Convert Assessments

Step 1: Go to the **Convert Assessment** tab.

Step 2: Upload one or more assessment .docx files (question papers, answer keys).

Step 3: Click **Convert**. The AI extracts questions, answers, and K/A mappings.

Step 4: Download the converted files as a ZIP — each file is in standard WSQ format.

10.2 Convert Courseware

Step 1: Go to the **Convert Courseware** tab.

Step 2: Upload AP, FG, LG, or ASR documents (.docx).

Step 3: Click **Convert** to reformat into WSQ standard.

10.3 Convert Lesson Plans

Step 1: Go to the **Convert Lesson Plan** tab.

Step 2: Upload existing lesson plan .docx files.

Step 3: Click **Convert** to restructure into the standard timetable format.

11. Courseware Audit

The audit tool cross-checks your generated courseware documents against the Course Proposal (source of truth) to catch any mismatches or errors.

11.1 Quick Check Mode

If you have just generated documents on the AP/FG/LG page, you can use **Quick Check**:

Step 1: Navigate to **Courseware Audit** in the sidebar.

Step 2: Click **Quick Check** — it automatically audits the documents you just generated using the session data.

Step 3: Review the audit table: green checkmarks (✓) for matches, red crosses (X) for mismatches.

11.2 Upload & Check Mode

For auditing existing documents (not just generated ones):

Step 1: Upload the **Course Proposal** (.docx, .xlsx, or .pdf).

Step 2: Upload the **courseware documents** to audit (AP, FG, LG, LP).

Step 3: Click **Audit**. The AI compares fields across all documents.

Step 4: Review the results table showing: field name, CP value, document value, and status.

Fields checked include:

- Course title, TSC code, TSC title
- Learning outcomes and topics
- Assessment methods and durations
- Organization name and UEN
- Proficiency levels and credit hours

12. Company Management (Settings)

Manage the training provider companies whose branding appears on generated documents.

12.1 Adding a Company

Step 1: Click the **Companies** button at the bottom of the sidebar.

Step 2: Click **Add New Company**.

Step 3: Fill in the details:

- **Company Name** — Full legal name
- **UEN** — Unique Entity Number (e.g., 20120096W)
- **Address** — Registered address
- **Email** — Contact email
- **Phone** — Contact number

Step 4: Upload the **company logo** (PNG or JPG). This appears on all documents.

Step 5: Click **Save**.

12.2 Editing / Deleting a Company

- **Edit:** Click the edit button next to any company to update its details or logo.
- **Delete:** Click delete — the system auto-backs up the company data before removing it (saved in company/deleted_companies/).
- **Search:** Use the search bar to find companies by name, UEN, or address.

***Note:** Always add a new client company BEFORE generating their courseware or brochure. The company logo, UEN, and address appear on the AP, FG, LG, Slides, and Brochure. If the company is missing from the database, these fields will be blank or show defaults.*

13. File Organisation & Output

13.1 Courseware Folder Structure

All generated documents are automatically saved to a **Courseware/** folder in the project directory. Each course gets its own subfolder, organised by document type:

Folder	Contents
Courseware/[Course Name]/	Root folder for each course
Course Proposal/	Original uploaded CP document
AP/	Assessment Plan and ASR documents
FG/	Facilitator Guide document
LG/	Learner Guide document
Lesson Plan/	Lesson Plan document
Assessment/	Question papers and answer keys
Slides/	PowerPoint slide deck
Brochure/	Marketing brochure PDF

13.2 File Naming Convention

All generated files follow a consistent naming pattern for easy identification:

Document	Filename Pattern	Example
Assessment Plan	Assessment_Plan_[TGS]_[Title].docx	Assessment_Plan_TGS-2024001234_Python.docx
ASR	Assessment_Summary_Record_[TGS]_[Title].docx	Assessment_Summary_Record_TGS-2024001234_Python.docx
Facilitator Guide	FG_[TGS]_[Title].docx	FG_TGS-2024001234_Python.docx
Learner Guide	LG_[TGS]_[Title].docx	LG_TGS-2024001234_Python.docx
Lesson Plan	LP_[TGS]_[Title]_v1.docx	LP_TGS-2024001234_Python_v1.docx
Assessment QP	[Type]_QP_[TGS]_[Title].docx	SAQ_QP_TGS-2024001234_Python.docx
Assessment AK	[Type]_AK_[TGS]_[Title].docx	SAQ_AK_TGS-2024001234_Python.docx

14. Troubleshooting & FAQ

Common Mistakes to Avoid

- **Skipping Extract Course Info** — Always start with Step 1. All other pages need this data.
- **Wrong company selected** — Check the sidebar dropdown before generating. The wrong company means wrong logo, UEN, and address on all documents.
- **Generating brochure without adding client company first** — Add the client in Company Management (Section 12) with their logo, address, and email before generating their brochure.
- **Opening DOCX in Google Docs** — Always use Microsoft Word. Google Docs may break the formatting of WSQ templates.
- **Not reviewing extracted data** — Always check the extracted course info before generating. If the AI misinterpreted something, the error will carry into all documents.

Problem	Solution
App won't load / page error	Refresh the page (F5 or Ctrl+R). If it still doesn't load, contact your administrator to check if the app is running.
'No course info loaded' warning	Go to Extract Course Info (page 1) and upload a Course Proposal first. All other pages depend on this.
Generation is stuck / spinning forever	Check the Agent Status in the sidebar. If a job is stuck, refresh the page (F5) and try generating again.
Generated document has wrong company name	Check the Company dropdown in the sidebar — make sure the correct company is selected before generating.
Downloaded DOCX looks wrong / formatting issues	Open the file in Microsoft Word (not Google Docs or LibreOffice) for best results. WSQ templates are optimized for MS Word.
Brochure generation fails	Ensure the Course URL is valid and accessible. Some websites block automated scraping.
Slides have fewer than expected	The system auto-pads to hit the target count. If still short, try regenerating.
Need to re-extract course info	Simply upload the CP again on the Extract Course Info page. It will overwrite the previous extraction.
Assessments not attaching to AP	Make sure you are uploading the correct AP file (generated from Section 4) and that the assessment files are .docx format.
Company logo not appearing on documents	Go to Company Management (Section 12) and check that a logo image (PNG/JPG) has been uploaded for the selected company.
Lesson Plan times don't add up	The Lesson Plan auto-fills gaps with breaks so every day is exactly 9am-6pm. If topics seem short, check the instructional hours in the CP.

Quick Reference — Recommended Workflow

Step	Page	Action	Output
1	Extract Course Info	Upload CP + enter TGS code	Extracted JSON (session)
2	Generate AP/FG/LG	Select docs → Generate	AP, ASR, FG, LG (.docx)
3	Generate Lesson Plan	Click Generate	LP (.docx)
4	Generate Assessment	Select types → Generate	Assessment papers (.docx)
5	Add Assessment to AP	Upload AP + assessments → Merge AP with annexes (.docx)	
6	Generate Slides	Click Generate	Slide deck (.pptx)
7	Generate Brochure	URL (Tertiary) or CP (Client)	Brochure (.pdf)
8	Courseware Audit	Quick Check or Upload	Audit report